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Acute Toxicity of Atrazine, Alachlor, and Chlorpyrifos Mixtures to Honey Bees

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Abstract

The acute toxicity of chlorpyrifos and chlorpyrifos-oxon (organophosphorothioate insecticides) was examined alone and in combination with atrazine (triazine herbicide) and alachlor (chloroacetanilide herbicide) to honey bees (*Apis mellifera*). Atrazine and alachlor were observed to not be acutely toxic to bees at doses up to 10 and 4 µg per bee, respectively. However, atrazine significantly increased chlorpyrifos toxicity by 3-fold while reducing chlorpyrifos-oxon toxicity by 1.8-fold. These changes in toxicity are correlated with significant 1.3- and 1.2-fold inhibition of acetylcholinesterase (AChE) activity in bees exposed to chlorpyrifos and chlorpyrifos-oxon, respectively. Atrazine significantly increased cytochrome P450, general esterase, and glutathione S-transferase (GST) activities by 1.5-, 1.2-, and 1.2-fold, respectively, in bees compared to untreated individuals. Alachlor increased chlorpyrifos toxicity by 2.5-fold but did not affect the toxicity of chlorpyrifos-oxon. Exposure to alachlor and chlorpyrifos did not affect AChE compared to chlorpyrifos alone. However, exposure to chlorpyrifos-oxon and alachlor significantly increased acetylcholinesterase (AChE) activity by 1.4-fold. GST activity, but not P450 or general esterases, was significantly increased in bees exposed to alachlor. These data provide evidence that triazine and chloroacetanilide herbicide exposure alters detoxification enzyme activity and, in turn, alters the sensitivity of bees to organophosphorothioate

insecticides. Importantly, these data can be used to guide future studies aiming to test safety profiles for pollinators and expand regulatory framework required for pesticide registration.

Keywords: pollinator, herbicide, insecticide, synergism, detoxification enzymes

1. Introduction

The western honey bee, *Apis mellifera*, is the most widely used crop pollinator species worldwide, enhancing the production of 35% of global crop species (Klein et al., 2007; Winfree et al., 2011; Delaplane et al., 2000; Free, 1970). These services contribute an estimated US\$20 billion to the US agricultural economy each year, and more than US\$230 billion globally (Gallai et al., 2009; Johnson, 2010). The increasing demand and cultivation of crops that require insects for pollination introduces a unique problem associated with significant albeit unintentional exposure to agrochemicals used by farmers to protect crops from pests and diseases (Mullin et al., 2010; Johnson et al., 2010; Meixner, 2010). The current regulatory framework requires pesticide registrants to conduct studies to assess the potential risk of pesticides to pollinators before advancing to a registered-use product (Gray, 2002; Panel, 2012). However, the current risk assessment guidelines focus on the risk of single active ingredients or formulated products without considering the cumulative risks from interactions of multiple pesticides (économiques, O.d.c.e.d.d., 1998, Test No. 213; OECD, 2017; OECD, 2013; OECD, 1998; OECD, 2014). Due to simultaneous exposure to multiple classes of crop protection products during honey bee pollination, it is important to understand the cumulative effects of common pesticides to bee survivorship that will provide a more complete understanding of pesticide exposure to pollinator health. Organophosphorothioate insecticides elicit toxicity by inhibiting the function of acetylcholinesterase (AChE), leading to increased synaptic acetylcholine concentrations and over-stimulation of cholinergic synapses, resulting in excitatory paralysis and death (Matsumura, 1985). The *in vitro* AChE potency and corresponding biological activity of thiophosphoryl (P=S) esters is low and, thus, bioactivation of organophosphorothioate insecticides is required *via* cytochrome P450 monooxygenase(s) to a phosphoryl (P=O) ester to elicit toxicity (Matsumura, 1985; Fukuto and Fahmy, 1981). Synergistic interactions between crop protection products are documented across several insect species (Thompson, 1996), and these data indicate an overall increase in toxicity after exposure to pesticide mixtures compared to exposure to individual pesticides (Anderson and Lydy, 2002; Anderson et al., 2008; Anderson and Zhu, 2004; Jin-Clark et al., 2008; Jin-Clark et al., 2002; Li et al., 2018; Zhu et al., 2014).

Chlorpyrifos (organophosphorothioate (OP) insecticide; AChE inhibitor), atrazine (triazine herbicide; photosynthesis inhibitor), and alachlor (chloroacetanilide herbicide; biosynthesis inhibitor) are crop protectants that have been widely used across multiple agricultural landscapes (Anderson and Zhu, 2004; Lichtenstein et al., 1973; Belden and Lydy, 2000; Ostiguy et al., 2019; Dietemann et al., 2013). Prior to the banning of chlorpyrifos on all food-based registrations in 2022 (EPA, 2022), chlorpyrifos was the most widely used insecticide in US agriculture (Atwood and Paisley-Jones, 2017) and was commonly used on bee-dependent crops, such as almond, alfalfa, and apple (Solomon et al., 2014). Although alachlor has been banned for use in the United States, atrazine is still widely used

on bee-relevant crops (Tavares et al., 2015; Company, 2018) and is the most abundant herbicide residue in bee colonies in both pollen and wax combs (Mullin et al., 2010). While chlorpyrifos and alachlor have been banned for use on food systems (Atwood and Paisley-Jones, 2017; EPA, 2016), other chloroacetanilides herbicides (e.g., metolachlor) and OPs (e.g., acephate, diazinon, malathion, etc.) are widely used in agricultural landscapes and have potential to negatively impact pollinator health, thus highlighting the need to test potential toxicological interactions between herbicides and OPs in honey bees.

Atrazine exposure has been documented to increase OP toxicity, reduce AChE activity, and increase detoxification enzyme activity in dipteran insects (Anderson and Zhu, 2004; Jin-Clark et al., 2002; Lichtenstein et al., 1973; Belden and Lydy, 2000). Similarly, alachlor exposure increased chlorpyrifos toxicity and reduced GST activity in dipteran insects when compared to exposure to chlorpyrifos only (Jin-Clark et al., 2008). The increased toxicity after co-exposure to atrazine and alachlor with OP insecticides is relevant to pollinators because these herbicides have been detected in the pollen and wax in managed bee hives (Ostiguy et al., 2019). The exposure of bees to atrazine and alachlor raises potential concern for the increased mortality of individual bees when exposed to chlorpyrifos during pollination events. Considering this, the goal of this study was to use toxicological and biochemical methods to test if atrazine and alachlor alter chlorpyrifos toxicity via modulation of detoxification enzyme activities in bees. We report the acute toxicity of chlorpyrifos alone and in combination with atrazine and alachlor to bees as well as these pesticides affecting AChE, general esterase, GST, and cytochrome P450 activities. The evaluation of triazine or chloroacetanilide herbicides as potential synergists of OP toxicity to honey bees may provide a framework for risk assessment of pesticide mixtures or simultaneous exposure by honey bees.

2. Materials and methods

2.1. Honey bees

Bee colonies of Italian stock were acquired from the USDA-ARS Honey Bee Breeding, Genetics, and Physiology Research Facility (Baton Rouge, Louisiana, USA). All colonies were maintained in standard Langstroth-style equipment. Commercially available Italian queens were purchased from C. F. Koehnen and Sons (Glenn, California, USA) and installed two months prior to the start of all experiments. Colonies were treated for *Varroa* mites prior to the study, and mite levels were monitored throughout the study at two-week intervals using the alcohol wash technique (Dietemann et al., 2013). Colonies with *Varroa* levels above 2 mites per 100 bees were excluded from the study.

2.2. Chemicals

Acetone (99.5% purity, ACS certified), chlorpyrifos (98.0%), alachlor and atrazine (PESTANAL analytical standard, $\geq 98\%$ purity) were purchased from Millipore-Sigma (St. Louis, Missouri, USA). Chlorpyrifos-oxon was purchased from ChemService (West Chester, Pennsylvania, USA). Acetylthiocholine iodide (ATChI), bicinehoninic acid solution, bovine serum albumin, 5,5-dithio-bis (2-nitrobenzoic acid) (DTNB), 7-ethoxycoumarin (7-EC), β -nicotinamide adenine dinucleotide phosphate (reduced β -NADPH), oxidized glutathione,

glutathione reductase, acetonitrile, TRIZMA-base, 7-ethoxycoumarin (7-EC), α -naphthyl acetate (α -NA), Triton X-100, fast blue B salt (*O*-dianisidine, tetrazotized), sodium dodecyl-sulfate (SDS), α -naphthol, glutathione, and 1-chloro-2, 4-dinitro-benzene (CDNB) were purchased from Sigma-Aldrich (St. Louis, Missouri, USA).

2.3. Acute toxicity bioassays

The acute toxicities of chlorpyrifos, chlorpyrifos-oxon, atrazine, and alachlor alone and in combination were determined using bees < 24 h post-emergence. Frames were removed from the hive, cleared of adult bees, and transferred to screened cages maintained in a dark incubator at 32.5 °C and 55%–65% relative humidity (Dietemann et al., 2013). After 24 h, newly emerged bees were collected and placed in standard bee study cages (Small-Life Supplies, Whittlesey, Peterborough, UK) and fed *ad libitum* 1:1 sucrose: water solution (OECD, 1998; Dietemann et al., 2013). To quantify changes in toxicity, dose response curves were constructed to determine the dose of chlorpyrifos and chlorpyrifos-oxon required to kill 25% of the test population (LD₂₅). The use of LD₂₅ was chosen to observe the maximal changes in toxicity of bees exposed to herbicides and to ensure a sufficient number of surviving bees for use in detoxification enzyme assays. Dose response curves were constructed by observing the mortality of bees exposed to 8 doses of chlorpyrifos (10, 30, 40, 45, 50, 60, 100, 300 ng/bee) and chlorpyrifos-oxon (5, 10, 15, 20, 30, 40, 50, 60 ng/bee), which resulted in a range of 0% to 100% mortality after topical application. Stock solutions of each insecticide were made in acetone and serially diluted for working solutions. Bees were anesthetized on ice for 3 min and 1 μ L of each insecticide solution was applied to the dorsal thorax of each bee using a 50 μ L Hamilton series 700 syringe and PB600–1 repeating applicator (Hamilton Company, Reno, Nevada, USA). Bee mortality for each dose was pooled from 4 replicated trials, where each trial consisted of 20 bees per cage and 3 cages per dose. The endpoint for each bioassay was measured as a lethal dose where mortality was defined as no visible movement upon probing with forceps during 5 s of observation. Mortality observations were performed 24 h after insecticide exposure. The LD₂₅ values were estimated using nonlinear regression analysis (GraphPad Prism 9, San Diego, California, USA).

Atrazine working solutions was prepared at 1, 5, and 10 μ g/bee in acetone. Alachlor working solutions were prepared at 4 μ g/bee (~10% of reported LD₅₀ (36 μ g/bee) (US-EPA, 1998), 2 and 1 μ g/bee. The US EPA Registration Eligibility Decisions were used to choose these concentrations of each herbicide (US-EPA, 1998; US-EPA, 2020). Atrazine was not observed to be lethal to bees at solubility limit (US-EPA, 2020), whereas alachlor has been shown to be lethal to bees at doses higher than those used for this study (US-EPA, 1998). The bees were topically exposed to herbicides as described above for the insecticides. Additionally, the bees were exposed to each dose of atrazine and alachlor alone and in combination with the LD₂₅ of chlorpyrifos and chlorpyrifos-oxon.

2.4. Acetylcholinesterase (AChE) activity

Acetylcholinesterase (AChE) activity of surviving bees exposed to chlorpyrifos, chlorpyrifos-oxon, atrazine, and alachlor alone and in combination, as previously described, was determined according to the modified method of Ellman et al. and Vu et al (Ellman et al., 1961; Vu et al., 2020). Briefly, each bee sample was homogenized and centrifuged with the

supernatant being used as the enzyme source for measuring AChE activity in a 96-well microplate containing ATCh and DTNB. The absorbance of the ATCh hydrolysis reaction was measured at 405 nm using a SpectraMax i3x multimode microplate reader (Molecular Devices, Inc., Sunnyvale, California). The results are reported as mean specific activity ($\mu\text{mol}/\text{min}/\text{mg}$ of total protein) $\pm$ standard error for each treatment.

2.5. General esterase activity

General esterase (GES) activity of surviving bees exposed to chlorpyrifos, chlorpyrifos-oxon, atrazine, and alachlor alone and in combination, as previously described, was determined according to the modified method of Vu et al. and O'Neal et al. using α -NA as the substrate (Vu et al., 2020; O'Neal et al., 2019). Briefly, each bee sample was homogenized and centrifuged with the supernatant being used as the enzyme source for measuring GES activity in a 96-well microplate containing α -NA and incubated at 37 °C for 30 min followed by the addition of fast blue B solution to stop the reaction. The absorbance of the GES hydrolysis reaction was measured at 600 nm using a SpectraMax i3x multimode microplate reader and the hydrolytic product was determined using an α -naphthol standard curve. The results are reported as mean specific activity ($\mu\text{mol}/\text{min}/\text{mg}$ of total protein) $\pm$ standard error for each treatment.

2.6. Glutathione S-transferase activity

Glutathione S-transferase (GST) activity of surviving bees exposed to chlorpyrifos, chlorpyrifos-oxon, atrazine, and alachlor alone and in combination, as previously described, was determined according to the modified method of Vu et al. and O'Neal et al. using CDNB as the substrate (Vu et al., 2020; O'Neal et al., 2019). Briefly, each bee sample was homogenized and centrifuged with the supernatant being used as the enzyme source for measuring GST activity in a 96-well microplate containing CDNB and glutathione. The absorbance of the GST conjugation reaction was measured at 340 nm using a SpectraMax i3x multimode microplate reader. The results are reported as mean specific activity ($\mu\text{mol}/\text{min}/\text{mg}$ of total protein) $\pm$ standard error for each treatment.

2.7. Cytochrome P450 activity

Cytochrome P450 activity of surviving bees exposed to chlorpyrifos, chlorpyrifos-oxon, atrazine, and alachlor alone and in combination, as previously described, was determined according to the modified method of Vu et al. (Vu et al., 2020) and O'Neal et al. (O'Neal et al., 2019) using 7-EC as a substrate following the preparation of each bee sample using a microsome isolation kit (Abcam, Cambridge, United Kingdom). Briefly, the resuspended microsome samples were transferred to a 96-well microplate containing 7-EC and reduced β -NADPH. The samples were incubated at 30 °C for 30 min followed by the addition of oxidized glutathione and glutathione reductase and then incubated at 37 °C for 15 min. The P450 O-deethylation reaction was stopped using acetonitrile in TRIZMA-base (pH 10), and the relative fluorescence units of the reaction were measured at 465 nm while exciting at 390 nm using a SpectraMax i3x multimode microplate reader. The results are reported as mean relative fluorescence (relative fluorescence (RF)/mg of total protein) $\pm$ standard error for each treatment.

2.8. Total protein content

Total protein in each sample preparation was determined using a bicinchoninic acid assay with bovine serum albumin as a standard (Vu et al., 2020; O'Neal et al., 2019). The total protein was measured at 560 nm using a SpectraMax i3x multimode microplate reader. The total protein content of each bee sample was used to correct the measures for AChE, P450, GES, and GST activity assays and standardize the measures per mg of total protein.

2.9. Statistical analysis

Dose-response curves to determine LD₅₀ values of organophosphates were calculated by nonlinear regression (variable slope) using GraphPad Prism (GraphPad Software, San Diego, California, USA). Each drug concentration was replicated 7–10 times. Differences in percent mortality during acute toxicity bioassays as well as specific enzyme activities based on treatments were statistically compared using a one-way analysis of variance (ANOVA) followed by a Fisher's least significant difference (LSD) test. Statistical differences were defined at a significance level (α) of 0.05.

3. Results

3.1. Toxicity bioassays with individual pesticides and binary mixtures

The LD₂₅ value for chlorpyrifos was determined to be 37.5 ng/bee (95% CI: 33–42 ng/bee, Hillslope: 11.9, r^2 : 0.95; Supplemental Fig. 1A). The LD₂₅ value for chlorpyrifos-oxon was estimated as 14 ng/bee (95% CI: 13–19 ng/bee, Hillslope: 11.8, r^2 : 0.91; Supplemental Fig. 1B), which is 2.7-fold more toxic than chlorpyrifos. Neither atrazine nor alachlor were acutely toxic to bees at the doses tested in this study (Fig. 1). The percent mortality for bees exposed to chlorpyrifos at the LD₂₅ in combination with 1-, 5-, and 10- μ g/bee atrazine was significantly increased by 2.8-, 2.8-, and 3.0-fold, respectively ($P < 0.05$), compared to individuals exposed to chlorpyrifos alone (Fig. 1A). The percent mortality for bees exposed to chlorpyrifos-oxon at the LD₂₅ in combination with 1, 5, and 10 μ g/bee atrazine was significantly reduced by 1.6-, 1.6-, and 1.8-fold ($P < 0.05$) compared to individuals exposed to chlorpyrifos-oxon alone (Fig. 1B).

The percent mortality for bees exposed to chlorpyrifos at the LD₂₅ in combination with 4 μ g/bee alachlor was significantly increased by 2.5-fold ($P < 0.05$) compared to individuals exposed to chlorpyrifos alone (Fig. 1C). However, the percent mortality for bees exposed to chlorpyrifos in combination with 2 μ g/bee alachlor was significantly reduced by 1.6-fold ($P < 0.05$) compared to individuals exposed to chlorpyrifos alone (Fig. 1B). The percent mortality for bees exposed to chlorpyrifos-oxon at the LD₂₅ in combination with 1 μ g/bee alachlor was significantly increased by 2.9-fold ($P < 0.05$) compared to individuals exposed to chlorpyrifos-oxon alone (Fig. 1D). Neither 2 μ g/bee nor 4 μ g/bee alachlor affected the percent mortality for bees exposed to chlorpyrifos-oxon (Fig. 1D).

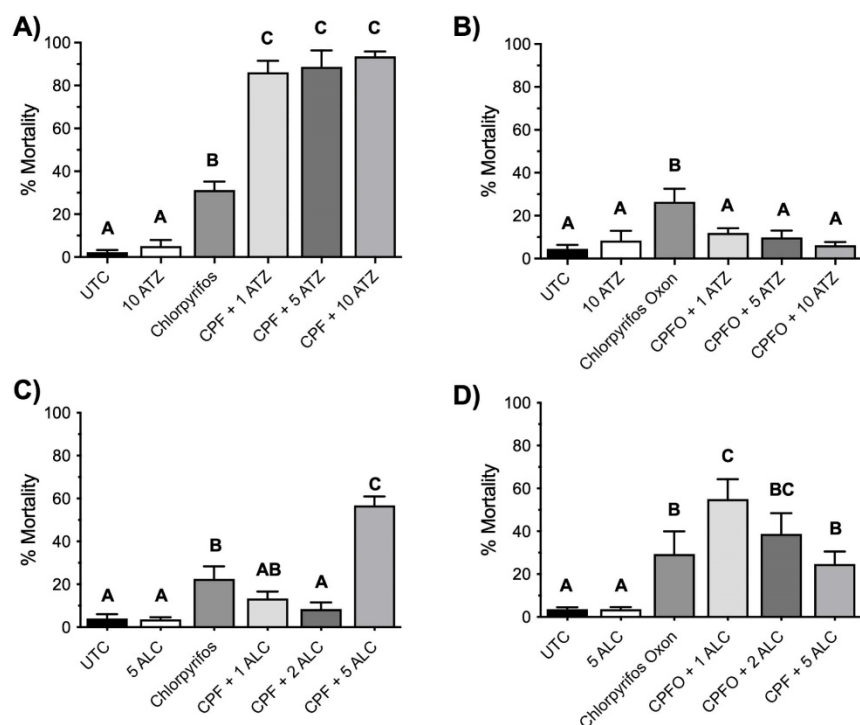


Figure 1. Acute toxicity of agrochemical mixtures to honey bees. Chlorpyrifos (A) or chlorpyrifos oxon (B) alone or in combination with atrazine was quantified with atrazine concentrations of 1-, 5-, or 10- $\mu\text{g}/\text{bee}$. Chlorpyrifos (C) or chlorpyrifos oxon (D) alone or in combination with alachlor was quantified with atrazine concentrations of 1-, 2-, or 5- $\mu\text{g}/\text{bee}$. Chlorpyrifos and chlorpyrifos oxon were studied at 37.5 ng/bee and 14 ng/bee, which is their respective LD_{25} . Acetone was used for the control treatment (UTC). Bars represent average ($n = 5$) percent mortality and error bars represent SEM. Bars not labeled by the same letter represents statistical significance at $P < 0.05$ as determined by one-way ANOVA with Fisher's post-hoc test.

3.2. Acetylcholinesterase (AChE) activity

The AChE activities of bees exposed to chlorpyrifos and chlorpyrifos-oxon at the LD_{25} alone and in combination with 10 $\mu\text{g}/\text{bee}$ atrazine are shown in Table 1. Exposure to atrazine led to significant increases in AChE activity relative to untreated controls and chlorpyrifos alone. Exposure to chlorpyrifos alone significantly reduced AChE activity compared to untreated bees and atrazine alone ($P < 0.05$, Table 1). In contrast, combined exposure to atrazine and chlorpyrifos significantly reduced AChE activity relative to chlorpyrifos alone, atrazine alone, and untreated bees ($P < 0.05$, Table 1). Exposure to chlorpyrifos-oxon did not alter AChE activity relative to untreated bees but was significantly lower than bees exposed to atrazine alone. Combined exposure to atrazine and chlorpyrifos-oxon significantly increased AChE activity compared to chlorpyrifos-oxon alone ($P < 0.05$), significantly decreased relative to atrazine alone ($P < 0.05$) and was not different than untreated bees (Table 1). The AChE activities of bees exposed to chlorpyrifos and chlorpyrifos-oxon

at the LD₂₅ alone and in combination with 4 µg/bee alachlor are shown in Table 2. Compared to untreated bees, chlorpyrifos exposure significantly decreased AChE activity ($P < 0.05$) while alachlor exposure did not alter AChE activity (Table 2). Combined exposure to chlorpyrifos and alachlor did not alter AChE activity relative to chlorpyrifos alone but significantly reduced activity relative to alachlor alone and untreated bees ($P < 0.05$, Table 2). Exposure to chlorpyrifos-oxon did not alter AChE activity relative to untreated bees but was significantly reduced compared to bees treated with alachlor alone ($P < 0.05$, Table 2). Combined exposure to alachlor and chlorpyrifos-oxon significantly increased AChE activity relative to chlorpyrifos-oxon alone ($P < 0.05$) but was not different than alachlor alone or untreated bees (Table 2).

Table 1. Effects of atrazine (ATZ; 10 µg/bee) alone and in combination with chlorpyrifos (CPF, 37.5 ng/bee) and chlorpyrifos-oxon (CPF, 15 ng/bee) on enzyme specific activities in bees. Variance is represented as standard error of the mean ($n = 10$). Different letters indicate significant differences between the means in the same column as determined by a one-way ANOVA and Fisher's LSD ($P < 0.05$). UTC: acetone, AChE: acetylcholinesterase, GES: general esterases, GST: glutathione S-transferases, P450: cytochrome P450 monooxygenases.

Pesticide	AChE*	GES [‡]	GST [^]	P450 [‡]
UTC	0.48 ± 0.03 AE	0.06 ± 0.001 AD	0.40 ± 0.007 A	15.59 ± 1.52 A
CPF	0.37 ± 0.02 B	0.05 ± 0.002 D	0.53 ± 0.007 B	17.16 ± 1.29 A
ATZ	0.56 ± 0.02 C	0.07 ± 0.006 B	0.46 ± 0.006 C	24.15 ± 0.77 B
CPF + ATZ	0.28 ± 0.01 D	0.07 ± 0.002 AB	0.52 ± 0.006 B	21.06 ± 0.90 BCD
OXN	0.43 ± 0.03 E	0.04 ± 0.004 C	0.39 ± 0.011 A	15.67 ± 0.81 A
ATZ + OXN	0.50 ± 0.01 A	0.07 ± 0.003 AB	0.48 ± 0.005 CD	20.65 ± 1.24 D

*µmol/min/mg

[‡]µmol/min/mg

[^]µmol/min/mg

[‡]RFU/10000/mg

Table 2. Effects of alachlor (ALC; 4 µg/bee) alone and in combination with chlorpyrifos (CPF, 37.5 ng/bee) and chlorpyrifos-oxon (CPF, 15 ng/bee) on acetylcholinesterase, general esterase, glutathione *S*-transferase, and cytochrome P450 activities in honey bees. Acetone was used for the untreated control treatment (UTC). Variance is represented as standard error of the mean ($n = 10$). Different letters indicate significant differences between the means in the same column as determined by a one-way analysis of variance and Fisher's LSD ($P < 0.05$). UTC: acetone, AChE: acetylcholinesterase, GES: general esterases, GST: glutathione *S*-transferases, P450: cytochrome P450 monooxygenases.

Pesticide	AChE*	GES [†]	GST [^]	P450 [‡]
UTC	0.48 ± 0.03 AE	0.06 ± 0.001 AB	0.40 ± 0.007 AD	15.59 ± 1.52 AB
CPF	0.37 ± 0.02 B	0.05 ± 0.002 BD	0.53 ± 0.007 B	17.16 ± 1.29 AB
ALC	0.56 ± 0.05 AD	0.06 ± 0.005 A	0.49 ± 0.007 C	13.09 ± 1.76 AB
CPF + ALC	0.37 ± 0.02 BC	0.05 ± 0.001 BC	0.42 ± 0.005 AE	17.35 ± 2.02 AB
OXN	0.43 ± 0.03 BCE	0.04 ± 0.004 C	0.39 ± 0.011 D	20.20 ± 4.30 A
OXN + ALC	0.58 ± 0.02 D	0.04 ± 0.007 CD	0.42 ± 0.005 E	9.62 ± 1.00 B

*µmol/min/mg

[†]µmol/min/mg

[^]µmol/min/mg

[‡]RFU/10000/mg

3.3. General esterase (GES) activity

The GES activities of bees exposed to chlorpyrifos and chlorpyrifos-oxon at the LD₂₅ alone and in combination with 10 µg/bee atrazine are shown in Table 1. Atrazine significantly increased GES activity compared to untreated bees and chlorpyrifos alone ($P < 0.05$, Table 1). Exposure to chlorpyrifos alone did not alter GES activity relative to untreated bees (Table 1). Combined exposure to chlorpyrifos and atrazine significantly increased GES activity relative to chlorpyrifos alone ($P < 0.05$) but did not alter activity compared to atrazine alone or untreated bees (Table 1). Chlorpyrifos-oxon exposure significantly reduced GES activity relative to atrazine and to untreated bees ($P < 0.05$, Table 1). Combined exposure to chlorpyrifos-oxon and atrazine significantly increased GES activity compared to chlorpyrifos-oxon alone ($P < 0.05$) but was not different than atrazine alone (Table 1). The GES activities of bees exposed to chlorpyrifos and chlorpyrifos-oxon at the LD₂₅ alone and in combination with 4 µg/bee alachlor are shown in Table 2. Alachlor exposure did not alter GES activity relative to untreated bees. Exposure to chlorpyrifos significantly decreased GES activity compared to alachlor ($P < 0.05$) but was not different than untreated bees (Table 2). Combined exposure to chlorpyrifos and alachlor did not alter GES activity compared to chlorpyrifos alone or untreated bees but was significantly reduced compared to alachlor alone (Table 2). Exposure to chlorpyrifos-oxon significantly reduced GES activity relative to alachlor alone and untreated bees ($P < 0.05$). Combined exposure to chlorpyrifos-oxon and alachlor did not alter GES activity compared to chlorpyrifos-oxon alone but led

to significantly decreased activity compared with alachlor alone and untreated bees ($P < 0.05$, Table 2).

3.4. *Glutathione s-transferase (GST) activity*

The GST activities of bees exposed to chlorpyrifos and chlorpyrifos-oxon at the LD₂₅ alone and in combination with 10 µg/bee atrazine are shown in Table 1. Exposure to atrazine significantly increased GST activity compared to untreated bees ($P < 0.05$), and chlorpyrifos exposure significantly increased GST activity relative to atrazine alone as well as untreated controls ($P < 0.05$, Table 1). Combined exposure to chlorpyrifos and atrazine did not alter GST activity relative to chlorpyrifos alone but was significantly increased compared to atrazine alone and untreated bees ($P < 0.05$, Table 1). Exposure to chlorpyrifos-oxon did not alter GST activity relative to controls but was significantly decreased compared to atrazine alone. Combined exposure to atrazine and chlorpyrifos-oxon led to GST activity that was significantly increased compared to chlorpyrifos-oxon alone and untreated bees ($P < 0.05$, Table 1) but was not different than atrazine alone. The GST activities of bees exposed to chlorpyrifos and chlorpyrifos-oxon at the LD₂₅ alone and in combination with 4 µg/bee alachlor are shown in Table 2. Exposure to alachlor alone significantly increased GST activity relative to untreated bees ($P < 0.05$), while exposure to chlorpyrifos alone increased GST activity relative to both alachlor alone and untreated bees ($P < 0.05$, Table 2). Combined exposure to alachlor and chlorpyrifos significantly reduced GST activity relative to alachlor alone and chlorpyrifos alone ($P < 0.05$) but was not different from untreated bees (Table 2). Chlorpyrifos-oxon exposure significantly reduced GST activity relative to alachlor alone ($P < 0.05$) but was not different than untreated bees (Table 2). Combined exposure to alachlor and chlorpyrifos-oxon was significantly increased compared to chlorpyrifos-oxon alone, yet significantly reduced compared to alachlor alone ($P < 0.05$, Table 2).

3.5. *Cytochrome P450 monooxygenase (P450) activity*

The P450 activities of bees exposed to chlorpyrifos and chlorpyrifos-oxon at the LD₂₅ alone and in combination with 10 µg/bee atrazine are shown in Table 1. Exposure to atrazine significantly increased P450 activity relative to chlorpyrifos alone and untreated bees ($P < 0.05$), while exposure to chlorpyrifos alone did not alter P450 activity relative to controls (Table 1). Combined exposure to atrazine and chlorpyrifos and atrazine significantly increased P450 activity compared to chlorpyrifos alone and untreated bees ($P < 0.05$) but was not different compared to atrazine alone (Table 1). Chlorpyrifos-oxon exposure did not alter P450 activity compared to untreated bees but was significantly reduced compared to atrazine alone. Combined exposure to chlorpyrifos-oxon and atrazine led to increased P450 activity compared to chlorpyrifos-oxon alone and untreated bees but was significantly lower than bees exposed to atrazine alone. The P450 activities of bees exposed to chlorpyrifos and chlorpyrifos-oxon at the LD₂₅ alone and in combination with 4 µg/bee alachlor are shown in Table 2. There were no differences in GST activity between untreated bees, bees exposed to alachlor alone, chlorpyrifos alone, or simultaneously exposed to chlorpyrifos and alachlor. Bees exposed to chlorpyrifos-oxon did not differ in GST activity compared to untreated bees or alachlor alone. Simultaneous exposure to chlorpyrifos-oxon and alachlor significantly decreased GST activity ($P < 0.05$) compared to chlorpyrifos-oxon alone.

4. Discussion

Commercial farmers must protect their crops from phytophagous and sap-feeding arthropods, infection with pathogens, and competition from noxious weeds to ensure high yields and to maximize economic returns (Atwood and Paisley-Jones, 2017; Pimentel et al., 1992). Application of various agrochemicals is therefore necessary to control these pest organisms (Pimentel et al., 1992) but complicates crop pollination services provided by managed colonies of honey bees (Gallai et al., 2009; Johnson, 2010; Caron and Connor, 2013). At the intersection of crop protection and crop pollination goals lies the unintentional exposure of managed crop pollinators to a diverse blend of agrochemical residues. Evaluating the combined risks these agrochemicals pose to pollinator health is thus of substantial relevance to the apiculture industry and to ensure sustainability of apiculture. While current risk assessment guidelines aim to shield nontarget organisms from deleterious effects from pesticide exposure (Gray, 2002; Panel, 2012), most of this regulatory framework is focused on a single active ingredient or a formulated product. The identification of herbicides, such as atrazine and alachlor, in the pollen and wax matrices of honey bee colonies (Ostiguy et al., 2019) combined with known exposure to OP insecticides during pollination events highlights the need to understand the combined effect of commercialized agrochemicals to managed crop pollinators. The high agricultural use and common detection of OPs within bee colonies (Mullin et al., 2010; Johnson et al., 2010; Atwood and Paisley-Jones, 2017) led us to test the effects of common herbicides to the toxicity of chlorpyrifos.

Atrazine is one of the most commonly used herbicides in the US, with ~64–74 million lbs. of active ingredient used between 2011 and 2012 (Atwood and Paisley-Jones, 2017). Bee exposure to atrazine during pollination service contracts is nearly certain due to use in almond orchards and other bee-reliant cropping systems (DeSimone et al., 2009). Further, while atrazine is not sprayed in almond orchards, it has been detected in orchard groundwater and likely enters the agroecoscape from applications to adjacent properties (DeSimone et al., 2009; Dubrovsky et al., 1995). Other herbicides, such as chloroacetanilides (e.g., alachlor, metolachlor, and butachlor), have been widely used in the US for broadleaf weed control with 6 million to 9 million lbs. estimated to have been used between 2000 and 2001 (US EPA, 2001) and are also likely encountered by bees conducting crop pollination. Furthermore, OPs are among the most commonly used insecticidal class in US agriculture (Atwood and Paisley-Jones, 2017) and are one of the most commonly detected insecticide residues in bee colonies. Taken together, simultaneous exposure to OP insecticides and triazine/chloroacetanilide herbicides by honey bees is likely and has the potential to negatively impact the health and productivity of managed bee colonies.

Our data indicate that exposure to alachlor and atrazine at 4 and 10 µg/bee, respectively, were nontoxic to bees, yet binary combinations of atrazine with chlorpyrifos or chlorpyrifos-oxon resulted in a significant increase or decrease in toxicity, respectively (Fig. 1A and Fig. 1C). This shift in toxicity was less clear for alachlor, but a significant increase in bee toxicity was observed after chlorpyrifos exposure. These data support previous studies that show OP toxicity by atrazine and subsequent detoxification of respective *o*-analogs occurs in the aquatic midge (*Chironomus tentans*) (Anderson et al., 2008; Anderson and Zhu, 2004; Jin-Clark et al., 2002; Belden and Lydy, 2000; Miota et al., 2000),

earthworm (Lydy and Linck, 2003), and amphipod species (Trimble and Lydy, 2006). Likewise, alachlor and metolachlor are known to enhance chlorpyrifos toxicity in midges (Jin-Clark et al., 2008). Our data support these previous data sets through atrazine-mediated changes in chlorpyrifos toxicity that are associated with decreased AChE activity (Table 1) and likely leads to enhanced toxification in bees. Further, chlorpyrifos-oxon combined with atrazine or alachlor increased AChE activity that is likely due to enhanced detoxification (Table 1 and Table 2). Importantly, exposure to neither atrazine nor alachlor alone led to altered toxicity or AChE activity that suggests atrazine alone is not an effective inhibitor of AChE and that the altered toxicity shown in binary combinations with chlorpyrifos and its oxon is likely the result of an indirect mechanism.

Synergistic toxification of chlorpyrifos and enhanced AChE inhibition by atrazine are attributed to increased bioactivation of organophosphorothioate molecules to oxon or sulfoxide analogs by cytochrome P450 monooxygenases (Anderson and Zhu, 2004; Jin-Clark et al., 2002; Belden and Lydy, 2000; Miota et al., 2000; Lydy and Linck, 2003; Trimble and Lydy, 2006). Parent organophosphorothioate compounds are relatively poor inhibitors of AChE and require bioactivation through oxidative desulfonation of the phosphorothioate residue to increase potency and toxicity (Matsumura, 1985; Fukuto and Fahmy, 1981). As a more potent electrophile, oxygen increases molecule affinity for the catalytic serine hydroxyl residue in the active site of AChE and, thus, increases the inhibition potency of the OP (Fukuto and Fahmy, 1981; Eto and Zweig, 2018; Fukuto, 1990; Jokanović, 2001). Likewise, atrazine (10 µg/bee) significantly increased bee P450 activity compared to chlorpyrifos alone and binary combinations with chlorpyrifos significantly increased AChE (Table 1). Further, P450 activity was significantly increased when bees were concurrently exposed to atrazine and chlorpyrifos-oxon, suggesting that altered chlorpyrifos toxicity is likely due to increased bioactivation of chlorpyrifos to chlorpyrifos-oxon and detoxification of chlorpyrifos-oxon to less-toxic metabolites (Table 1). In contrast, atrazine exposure slightly but significantly increased GES activity relative to untreated control, and combined exposure to atrazine and chlorpyrifos or atrazine and chlorpyrifos oxon increased GES activity compared to the OPs alone or untreated control (Table 1). Similarly, atrazine alone or in combination with chlorpyrifos or chlorpyrifos oxon significantly increased GST activity (Table 1). Increases in either GES or GST activity would be expected to lead to increased detoxification, yet our data show that combined exposure to chlorpyrifos and atrazine leads to increased toxicity (Fig. 1A). The hypothetical decrease in toxicity due to increased GES or GST activity thus may be overshadowed by P450-mediated oxidative activation of chlorpyrifos-to-chlorpyrifos oxon (Table 1). When bees were exposed to chlorpyrifos oxon and atrazine, however, no further oxidative activation was possible, and increased GES and GST activity may be responsible the decrease in chlorpyrifos oxon toxicity (Table 1, Fig. 1B). Based on these data, we propose that atrazine alters the toxicity of chlorpyrifos through increased oxidative activation of the parent compound chlorpyrifos to increase AChE inhibition and toxicity. Reciprocally, chlorpyrifos-oxon cannot be further activated and thus undergoes increased detoxification, resulting in reduced AChE inhibition and decreased toxicity. These data support our central hypothesis that common crop protection products may synergistically enhance the toxicity of organophosphorothioate insecticides commonly used in US agriculture (Atwood and Paisley-Jones, 2017).

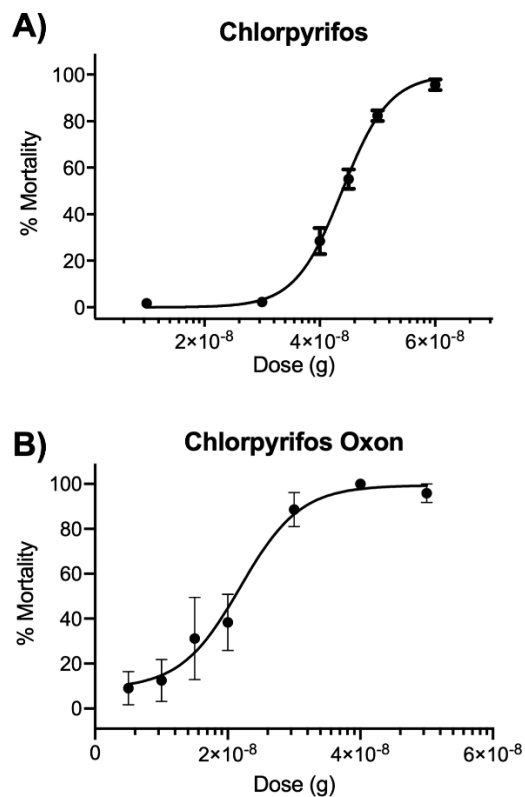
Our data provide evidence that alachlor exposure enhances chlorpyrifos toxicity (Fig. 1C) but does not alter AChE, GES, or P450 activity relative to controls or OPs alone (Table 2), indicating that other enzymes may be responsible for this change in toxicity. The activity of GSTs was significantly decreased upon exposure to alachlor alone and in combination with chlorpyrifos (Table 2), which has also been documented to occur in the aquatic midge (Jin-Clark et al., 2008). These authors further demonstrated reduced protein levels in midges exposed to metolachlor, another commonly used chloroacetanilide herbicide, which has been predicted to inhibit GST and other detoxification enzyme synthesis through inhibition of protein synthesis (Jin-Clark et al., 2008). Considering this, alachlor exposure in bees may reduce GST production that results in enhanced chlorpyrifos toxicity and subsequently increased inhibition of AChE in the absence of altered P450 activity. As observed with atrazine, these data further support the notion that common agrochemicals synergistically increase chlorpyrifos toxicity.

In conclusion, this study demonstrates that atrazine exposure significantly increases toxicity and AChE inhibition of the organophosphorothioate insecticide chlorpyrifos in honey bees. Although other enzymatic processes may be involved, the lack of effect of atrazine to AChE activity suggests the induction of cytochrome P450s leads to increased toxicity of chlorpyrifos when coexposed with atrazine. In contrast, alachlor increases chlorpyrifos toxicity without altering P450 levels, yet reduced GST activity (Table 2) in alachlor-exposed bees led us to speculate that the increased toxicity is partly due to decreased GST-mediated metabolism. Taken together, these data increase our understanding of agrochemical interactions to bee survivorship and can be used to guide agrochemical use patterns that reduce negative consequences to bee sustainability.

Data availability – Data will be made available on request.

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Appendix A. Supplementary data



Supplemental Figure 1. Dose response curves representing percent mortality of adult bees 24 hours after topical exposure to chlorpyrifos (**A**) and chlorpyrifos-oxon (**B**).

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